

MODULE 3: SENSORS AND INTERFACING

➤ SENSORS AND ACTUATORS

- An embedded system is in constant interaction with the Real world and the controlling/monitoring functions executed by the embedded system is achieved in accordance with the changes happening to the Real world.
- The changes in system environment or variables are detected by the sensors connected to the input port of the embedded system. If the embedded system is designed for any controlling purpose, the system will produce some changes in the controlling variable to bring the controlled variable to the desired value. It is achieved through an actuator connected to the output port of the embedded system.
- If the embedded system is designed for monitoring purpose only, then there is no need for including an actuator in the system. For example, take the case of an ECG machine. It is designed to monitor the heart beat status of a patient and it cannot impose a control over the patient's heart beat and its order. The sensors used here are the different electrode sets connected to the body of the patient. The variations are captured and presented to the user (may be a doctor) through a visual display or some printed chart.

➤ Sensors

A sensor is a transducer device that converts energy from one form to another for any measurement or control purpose.

Eg: the sensor which measures the distance between the cushion and magnet in the smart running shoe is a magnetic hall effect sensor.

➤ Actuators

Actuator is a form of transducer device (mechanical or electrical) which converts signals to corresponding physical action (motion). Actuator acts as an output device.

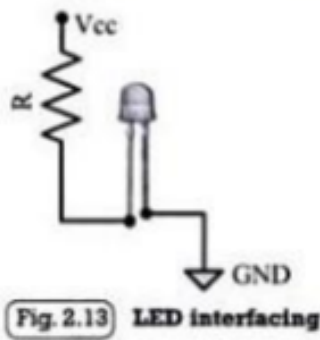
Eg: In the smart running shoes, the actuator used for adjusting the position of the cushioning element is a micro stepper motor

➤ The I/O Subsystem

- The I/O subsystem of the embedded system facilitates the interaction of the embedded system with the external world.
- The interaction happens through the sensors and actuators connected to the input and output ports respectively of the embedded system.
- The sensors may not be directly interfaced to the input ports, instead they may be interfaced through signal conditioning and translating systems like ADC, optocouplers, etc

➤ Light Emitting Diode (LED)

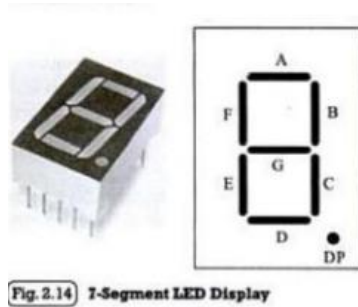
- Light Emitting Diode (LED) is an important output device for visual indication in any embedded system. LED can be used as an indicator for the status of various signals or situations.
- Typical examples are indicating the presence of power conditions like 'Device ON', 'Battery low' or 'Charging of battery' for a battery operated handheld embedded devices.
- Light Emitting Diode is a p-n junction diode which emits light when forward biased. It contains an anode and a cathode.
- For proper functioning of the LED, the anode should be connected to positive terminal of the supply voltage and cathode to the negative terminal of supply voltage.
- The current flowing through the LED must be limited to a value below the maximum current that it can conduct. A resistor is used in series between the power supply and the LED to limit the current through the LED.
- The ideal LED interfacing circuit is shown in Fig. 2.13.



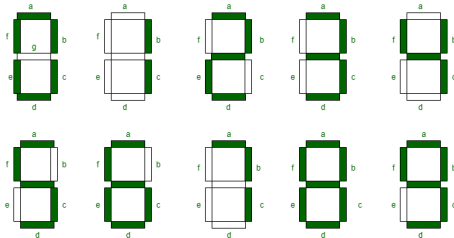
- LEDs can be interfaced to the port pin of a processor/controller in two ways.
- In the first method, the anode is directly connected to the port pin and the port pin drives the LED. In this approach the port pin 'sources' current to the LED when the port pin is at logic High (Logic '1').
- In the second method, the cathode of the LED is connected to the port pin of the processor/controller and the anode to the supply voltage through a current limiting resistor. The LED is turned on when the port pin is at logic Low (Logic '0'). Here the port pin 'sinks' current.
- If the LED is directly connected to the port pin, depending on the maximum current that a port pin can source, the brightness of LED may not be to the required level. In the second approach, the current is directly sourced by the power supply and the port pin acts as the sink for current. Here we will get the required brightness for the LED.

➤ 7-Segment LED Display

- The 7-segment LED display is an output device for displaying alpha numeric characters.
- It contains 8 light-emitting diode (LED) segments arranged in a special form. Out of the 8 LED segments, 7 are used for displaying alpha numeric characters (0 to 9 , A to F) and 1 is used for representing 'decimal point / dot point' in decimal number display. (For example to display 2.5, dot is used to represent the decimal point in this numeral.)
- Figure 2.14 explains the arrangement of LED segments in a 7-segment LED display.



- The LED segments are named A to G and the decimal point LED segment is named as DP. The LED segments A to G and DP should be lit accordingly to display numbers and characters.
- For example, for displaying the number 4, the segments F, G, B and C are lit.
- For displaying the character 'd', the segments B, C, D, E and G are lit.



- All these 8 LED segments need to be connected to one of the port pins of the processor/controller for displaying alpha numeric digits.
- The 7-segment LED displays are available in two different configurations, namely; **Common Anode and Common Cathode**.
- In the **common anode configuration**, the anodes of the 8 segments are connected commonly whereas **in the common cathode configuration**, the 8 LED segments share a common cathode line. **Figure 2.15 illustrates the Common Anode and Cathode configurations.**

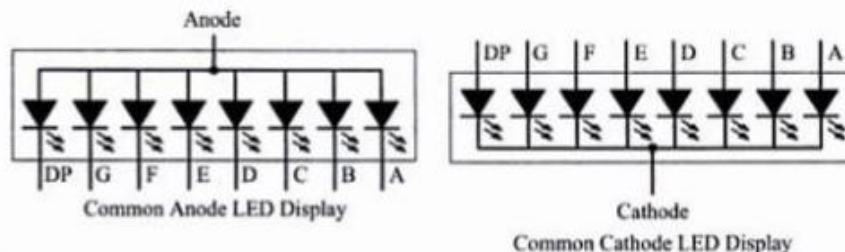


Fig. 2.15 Common anode and cathode configurations of a 7-segment LED Display

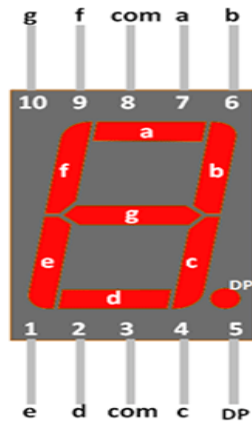
- Based on the configuration of the 7-segment LED unit, the LED segment's anode or cathode is connected to the port of the processor/controller in the order 'A' segment to the least significant port pin and DP segment to the most significant port pin.
- The current flow through each of the LED segments should be limited to the maximum value supported by the LED display unit.
- The typical value for the current falls within the range of 20mA. The current through each segment can be limited by connecting a current limiting resistor to the anode or cathode of each segment. The value for the current limiting resistors can be calculated using the current value from the electrical parameter listing of the LED display.
- For **common cathode configurations**, the anode of each LED segment is connected to the port pins of the port to which the display is interfaced. Here logic 0 (logic low ,normally Ground) is applied to the common cathode pin and logic high to the remaining pins.
- The truth table of Common Cathode 7 segment display for decimal digits

Segments Inputs							7 Segment Display Output
a	b	c	d	e	f	g	
1	1	1	1	1	1	0	0
0	1	1	0	0	0	0	1
1	1	0	1	1	0	1	2
1	1	1	1	0	0	1	3
0	1	1	0	0	1	1	4
1	0	1	1	0	1	1	5
1	0	1	1	1	1	1	6
1	1	1	0	0	0	0	7
1	1	1	1	1	1	1	8
1	1	1	1	0	0	1	9

- The anode of the **common anode LED display** is connected to the 5V supply voltage through a current limiting resistor and the cathode of each LED segment is connected to the respective port pin lines. Thus, in order to glow the LED, these cathodes have to be connected to the logic '0' and the common anode pin to the logic '1' (logic high, normally +5 voltage).
- The truth table of Common Anode 7 segment display for decimal digits

Segments Inputs							7 Segment Display Output
a	b	c	d	e	f	g	
0	0	0	0	0	0	1	0
1	0	0	1	1	1	1	1
0	0	1	0	0	1	0	2
0	0	0	0	1	1	0	3
1	0	0	1	1	0	0	4
0	1	0	0	1	0	0	5
0	1	0	0	0	0	0	6
0	0	0	1	1	1	1	7
0	0	0	0	0	0	0	8
0	0	0	0	1	1	0	9

- For an LED segment to lit in the Common cathode LED configuration, the port pin to which the anode of the LED segment is connected should be set at logic 1.
- For an LED segment to lit in the Common anode LED configuration, the port pin to which the cathode of the LED segment is connected should be set at logic 0.
- 7- Segment Display Pinout Diagram is shown below



- The pins 8 and 3 are shorted together and connected to ground (logic 0) in the case of Common cathode display or connected to Power (+5V or logic 1) if the display is of common anode type
- 7-segment LED display is a popular choice for low cost embedded applications like Public telephone call monitoring devices etc
- **Applications of 7 Segment Display:**
- Digital clocks
- Calculators
- Wrist watches
- Speedometers
- Radiofrequency indicators etc.